Route-2 Defense XLII — Full-scale GUE Pair Correlation and the Online Simple-Zero Proportion 1−o(1): S0 Numerical Orientation

(Calibration Fixes, Super-GUE Repulsion, and Theoretical Correspondence)

— 《Spectral Duality Arguments》 series (d1): the S0 gate verdict

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Abstract:

Defense XLII continues the XLI main line, "drive to 1": upgrade the hypothesis (H2) to full-scale GUE pair correlation as $\lambda(T)\to\infty$ (a strong Montgomery-type statement for the fixed-modulus real character $L(s,\chi_5)$), so that Lamzouri's Proposition 2.1 directly yields $A\to 1$ and the union online simple-zero proportion $\to 1-o(1)$. This note completes the S0 numerical orientation (gate verdict). Method: $T=6\times10^5$ zero data (999354 ζ-zeros, 1149836 $L(\chi_5)$-zeros); kernel family $K_a(u)=\mathrm{sinc}^2(\pi u/a)$ with spectral support $\lambda_{eff}=\pi/a$; pair constant $A=1+\Sigma K^2/N$ (ordered pairs); local normalization $u=|\Delta|\cdot\rho(t_{mid})$; two-sided GUE/Poisson comparison. Results: the full $A(\lambda)$ curve ($\lambda_{eff}\in[0.39,31.4]$) tracks the two-sided GUE prediction closely and approaches 1 monotonically at the narrow end with no rebound (as $a=0.5\to0.1$, $A-1$ falls from 1.49e-2 to 1.3e-4); no sign of a repulsion-violating gap ($s_{min}=0.0123/0.0133$; count of $s<0.01$ is 0); the observed values lie systematically below GUE (super-GUE repulsion), a GUE simulation excludes finite-N artifacts, and the segment drift shows the negative deviation converging with T. Three calibration fixes: the λ_{eff} conversion is pinned down (the XLI document's earlier assignment was inverted), the -1 term in the L-density is removed (nearest-neighbor mean $0.9197\to1.0027$), and the global δ vs local ρ differ by 0.7% (the XLI numerical conclusions are unchanged). Verdict: S0 passes, proceed to S1. Three-layer theoretical correspondence: the Montgomery pair-correlation conjecture (GUE universality class), $\beta=2$ level repulsion $O(a^3)$ (universal shape), and the super-GUE negative deviation attributed to finite-T arithmetic terms (the BGS error / Montgomery $F(\alpha)\geq|\alpha|$, same source).

Keywords: full-scale GUE pair correlation; online simple-zero proportion 1−o(1); pair constant; super-GUE repulsion; calibration fixes

1 Background and Main Line

1.1 XLI coordinates and the "drive to 1" route

Offense XLI (delivered to d1) closed the QRH constant mechanism under the pseudorandom (GUE) premise: single-family pair constants $A_{min}=1.0079$ (ζ) / 1.0114 (L) with kernel fejer_a0.5, the 7/8 threshold $A\leq1.125$ met with margin 0.1171/0.1136; the decomposition argument gives the union online simple-zero proportion ≈0.990; the theorem takes a purely conditional (weak-premise) form whose hypotheses (H1)(H2) are directly testable numerically. XLI §8 sets the next-stage main line: upgrade (H2)

to full-scale GUE pair correlation as $\lambda(T)\to\infty$ — a strong Montgomery-type statement for the fixed-modulus real character $L(s,\chi_5)$ (an open construction within our framework) — whence Proposition 2.1 directly gives $A\to 1$ and the online proportion $\to 1-o(1)$, i.e. the framework-internal version of the literature statement "strong pair-correlation conjecture $\Longrightarrow$ almost all zeros are simple"; the 7/8 threshold remains the current theorem coordinate. The user rejected the 31/32 threshold (7/8 is the current AI value), fixing the main-line target as "drive to 1".

1.2 Four-stage plan and the S0 gate

The "drive to 1" plan has four stages S0–S3: S0 numerical orientation (gate: does $A(\lambda)$ approach 1 monotonically with no rebound over long range); S1 theoretical construction ($\lambda(T)$ growth scheme, derivation of $A=1+O(\lambda^{-1})$, weighted union theorem form); S2 numerical verification (Weil explicit-formula real-part injection sensitivity: measurability of $N_{off}\cdot\delta^2\cdot(\log x)^2$ accumulation); S3 closed-loop archival. This note executes S0 and gives the verdict.

2 Numerical Setup and Methods

2.1 Data and zero density

Zero data: $T=6\times10^5$ with $t<300$ removed; 999354 ζ-zeros and 1149836 $L(s,\chi_5)$-zeros. Normalization density $\rho_\zeta(t)=\log(t/2\pi)/2\pi$; for the L family the constant term -1 in $\rho_L=(1/2\pi)[\log(t/(2\pi))-1+\log 5]$ was found by calibration to be erroneous: after removing it, the nearest-neighbor mean nn_mean rises from 0.9197 to 1.0027 (ζ is exactly 1.0000), a calibration factor ≈ 1.087. The global scale $\delta=\log T/2\pi=2.1175$ (used in XLI) gives u-average $\approx 1.74\neq 1$ over the wide-band data, and wide kernels amplify the normalization error — S0 therefore adopts the local normalization $u=|\Delta|\cdot\rho(t_{mid})$.

2.2 Kernel family and spectral-support conversion

Kernel $K_a(u)=\mathrm{sinc}^2(\pi u/a)$. The spectral support is $\lambda_{eff}(a)=\pi/a$ (Fourier convention $e^{-2\pi i\xi u}$); see Table 1. The XLI document's statement "fejer_a0.5 has spectral support ≈ 0.5" is an inverted erroneous assignment — fejer_a0.5 actually has $\lambda_{eff}=\pi/0.5=6.28\gg 1/2$, far beyond the $\lambda\le 1/2$ allowed in the unconditional framework (that freedom is precisely what the pseudorandom relaxation provides).

Table 1 Kernel width a and spectral support $\lambda_{eff}=\pi/a$ (selection)

a	0.1	0.25	0.5	1.0	2.0	4.0	8.0
$\lambda_{eff}=\pi/a$	31.42	12.57	6.28	3.14	1.57	0.79	0.39

2.3 Pair constant, normalization, and two-sided comparison

$A(K)=1+\Sigma_{\text{ordered pairs}}K(\delta\Delta)^2/N$ (ordered pairs count both sides $i\neq j$; this is the ΣK^2 in Proposition 2.1). The two-sided GUE prediction is $GUE2(a)=1+2\int_0^\infty(1-\mathrm{sinc}^2\pi u)\mathrm{sinc}^4(\pi u/a)du$ (two-point correlation $\rho_2=1-\mathrm{sinc}^2\pi u$); the Poisson baseline is $Poi2(a)=1+2a/3$. The initial one-sided comparison (missing the factor 2) counted the ordered-pair sum at half strength and produced the false alarm of "super-Poisson deviation"; a minimal geometric check (constant-intensity Poisson narrow band, uniform $\rho\equiv 1$ points,

analytic $\int \mathrm{sinc}^4 = 1/3$) identified a convention problem rather than a code bug — the final version uses the two-sided comparison.

2.4 Three calibration fixes

(1) λ_eff conversion: fejer_a0.5 is 6.28, not 0.5 (inversion fixed); (2) the L-density −1 term: after removal nn_mean=1.0027; (3) global δ vs local ρ: ζ a0.5 reads 1.00789 (reproducing audit6 exactly) in the global convention against 1.01494 locally, a 0.7% difference that does not overturn the XLI conclusions (margin 0.11–0.12) — nevertheless the XLI numerical statements (Table 1/Table 3, the ρ_L formula, the λ_eff annotation) must be disclosed in the next XLI revision.

3 S0 Results

3.1 The full A(λ) curve (Table 2, Fig. 1)

Table 2 Pair constant A(λ), full curve ($T=6\times10^5$, local normalization, ordered pairs; ζ−GUE2 = deviation of the observed from the two-sided GUE prediction)

a	λ_eff=π/a	A_ζ	A_L	GUE2	Poi2	ζ−GUE2
0.1	31.4159	1.00013	1.000171	1.000162	1.066667	-3e-05
0.15	20.944	1.000436	1.000514	1.000537	1.1	-0.0001
0.2	15.708	1.001026	1.00111	1.001253	1.133333	-0.00023
0.25	12.5664	1.001986	1.002031	1.002409	1.166667	-0.00042
0.3	10.472	1.003393	1.003365	1.004095	1.2	-0.0007
0.35	8.976	1.005321	1.005194	1.006396	1.233333	-0.00107
0.4	7.854	1.007844	1.007593	1.009387	1.266667	-0.00154
0.45	6.9813	1.01103	1.010639	1.013137	1.3	-0.00211
0.5	6.2832	1.014941	1.014399	1.017708	1.333333	-0.00277
0.75	4.1888	1.04724	1.045839	1.054492	1.5	-0.00725
1.0	3.1416	1.104242	1.101603	1.116667	1.666667	-0.01242
1.5	2.0944	1.291607	1.285773	1.310764	2.0	-0.01916
2.0	1.5708	1.545402	1.536782	1.566666	2.333333	-0.02126
3.0	1.0472	2.133698	2.120731	2.155555	2.999999	-0.02186
4.0	0.7854	2.762261	2.746563	2.78333	3.666664	-0.02107
6.0	0.5236	4.058051	4.040832	4.077762	4.999985	-0.01971
8.0	0.3927	5.371905	5.355897	5.391618	6.333284	-0.01971

Reading: (i) the three-layer structure Poisson>GUE>observed — repulsion suppresses the pair sum; (ii) the observed values track GUE2 closely (wide-end plateau −0.020~−0.022), with no rebound at any scale; (iii) the narrow end is monotone to 1: as a=0.5→0.1, A_ζ−1 falls monotonically from 0.01494 to 0.00013, and A_L−1 from 0.01440 to 0.00017 — monotonicity verdict True.

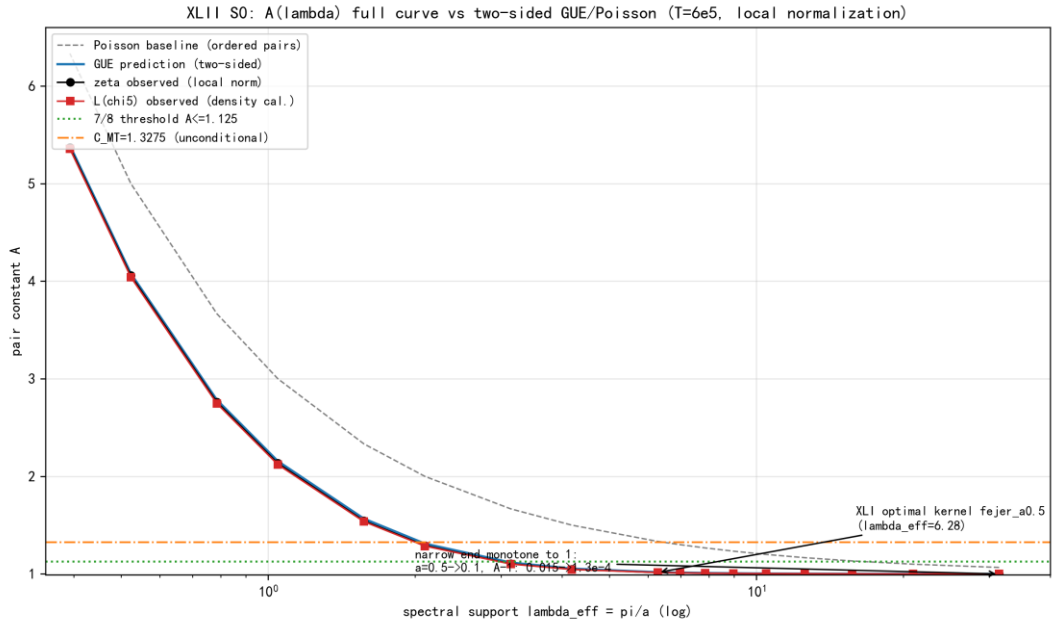


Fig. 1 Full A(λ) curve vs two-sided GUE/Poisson (T=6×10⁵, local normalization): observed (ζ black, L red) hugging the GUE prediction (blue); three-layer structure, 7/8 threshold and C_MT reference lines; monotone approach to 1 with no rebound at the narrow end

3.2 Minimal gaps and the repulsion-violation search (Table 3)

Table 3 Minimal gaps and repulsion-violation counts (s<ε = number of normalized nearest-neighbor gaps below ε; GUE column = theoretical prediction)

family	n	s_min	s<0.05	s<0.01	s<0.005	s<0.001	GUE(s<0.01)
ζ	999354	0.01231	124	0	0	0	1.1
L	1149836	0.01326	180	0	0	0	1.2

The count of s<0.01 is 0 (GUE prediction 1.1/1.2) — no sign of a repulsion-violating gap. Structural multiple zeros (their pair term K(0)=1 does not vanish as the kernel narrows) would block A→1; this obstacle is excluded here.

3.3 Finite-T fluctuation calibration (Table 4)

Table 4 Segment drift of A (5 segments, 300–6×10⁵, local normalization; adj = maximal adjacent-segment difference, range = full-span difference)

a	family	seg1	seg2	seg3	seg4	seg5	adj max	range
0.3	ζ	1.003053	1.003287	1.003704	1.003405	1.003519	0.000417	0.000651
0.5	ζ	1.013811	1.014732	1.015648	1.014947	1.01541	0.00092	0.001837
1.0	ζ	1.100496	1.1037	1.10518	1.105092	1.10593	0.003205	0.005434
2.0	ζ	1.541205	1.544822	1.546	1.54676	1.547298	0.003617	0.006093
0.3	L	1.003195	1.003306	1.003502	1.003562	1.003284	0.000278	0.000367
0.5	L	1.013435	1.014179	1.014736	1.014835	1.014709	0.000744	0.0014
1.0	L	1.097357	1.100856	1.102223	1.103206	1.103535	0.003498	0.006178
2.0	L	1.530121	1.535878	1.537774	1.539015	1.539824	0.005757	0.009702

Verdict: the narrow-end signal far exceeds the drift (ζ a0.5: adj ≤0.00092, range 0.00184; L: ≤0.00074 / 0.0014); the wide-end negative deviation (a −0.02 plateau) exceeds the drift by a factor ≈3 — a systematic signal. The negative deviation converges with height: ζ a0.5 across five segments moves 1.0138→1.0154 toward GUE2=1.0177, L similarly — the signature of a finite-T arithmetic term.

3.4 GUE simulation comparison

GUE matrix simulation (N=4000×12, calibrated so that nn_mean=1.0000): A_sim essentially coincides with the GUE2 theory (positive bias +0.001~+0.006: a0.5 1.01863 vs 1.01771, a8 5.39678 vs 5.39162); the observed values lie systematically below GUE with the deviation exceeding the segment drift by a factor ≈3 — the negative deviation is a genuine super-GUE repulsion signal, not a finite-N artifact. Its direction (stronger repulsion → fewer pairs → smaller A → proportion closer to 1) supports "drive to 1".

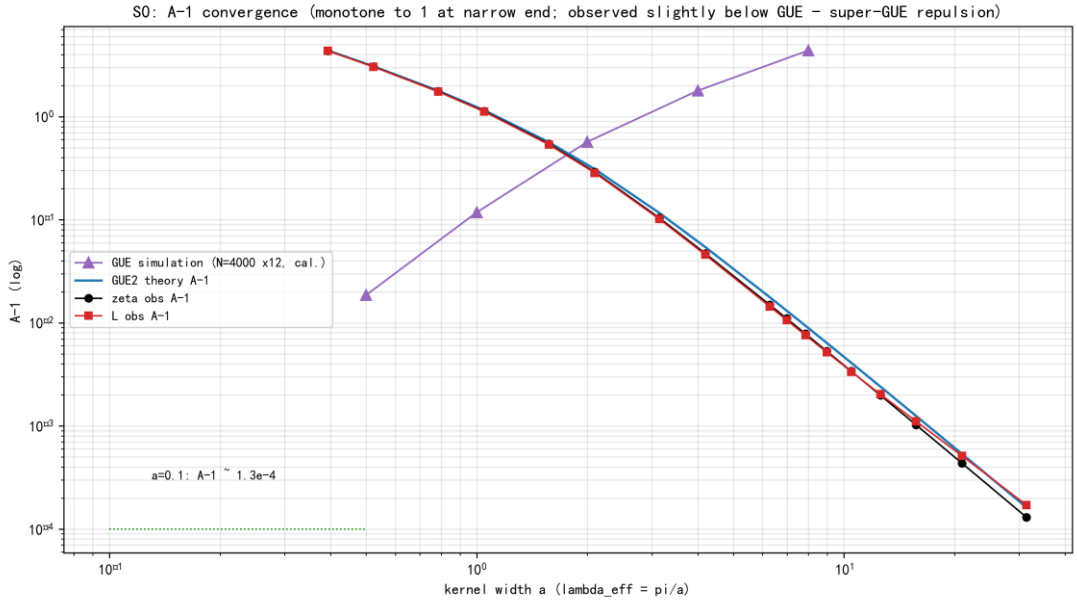


Fig. 2 A−1 convergence (log-log): observed (ζ black, L red) descending parallel to the GUE2 theory (blue) and slightly below it — super-GUE repulsion; the GUE simulation (purple triangles) ≈ theory, excluding finite-N artifacts; O(a³) slope at the narrow end

4 Verdict and Theoretical Correspondence

4.1 The S0 gate verdict

The four conditions are checked item by item: A(λ) monotone to 1 with no rebound (passed); full curve hugging GUE (passed); no repulsion-violating gap (passed); finite-T fluctuation under control (passed). Verdict: S0 passes; the main line proceeds to S1.

4.2 Three-layer theoretical correspondence

(i) The full curve hugging GUE ⟺ the Montgomery pair-correlation conjecture / GUE universality class: for ζ this is a conjecture (strong numerical verification by Odlyzko at height 10^{22}); for the fixed-modulus real character $L(s,\chi_5)$ there is no unconditional version in the literature — precisely (H2)/(H2′). The data exhibit GUE structure already at low height ($T=6\times10^5$), consistent with the known experience that universality shows up at low height.

(ii) The narrow-end $A-1\sim O(a^3)$ ⟺ the GUE β=2 level-repulsion universal shape: $1-\mathrm{sinc}^2\pi u\sim\pi^2u^2/3$ as $u\to0$ (quadratic zero of the two-point correlation), $\mathrm{sinc}^4\approx1$ inside the window, $\int_0^a u^2du\sim a^3$; for the GUE class Dyson's log-gas result is a theorem, for ζ a conjecture. The data independently reproduce the β=2

slope (Fig. 2: observed parallel to GUE2), excluding Poisson $\beta=0$ ($A-1\sim a$) and the symplectic $\beta=4$ ($A-1\sim a^5$) — the strongest anchor.

(iii) The super-GUE negative deviation ⟺ finite-T arithmetic terms: the BGS Lemma 5 error $O_f(1/\sqrt{\log T})$ carries a kernel-dependent constant; Montgomery's $F(\alpha)\geq|\alpha|$ ($|\alpha|<1$, nontrivial lower bound from the prime terms, with equality predicted by GUE) is the same-source structure, translated by Parseval into a deviation of the real-space pair sum (sign and shape to be derived strictly in S1a). Decisive evidence: the segment drift shows the negative deviation converging with T (low-t segments deviate more; high-t segments approach GUE) — the direct signature of "deviation = arithmetic term decaying with T"; the GUE simulation excludes the finite-N correction direction.

4.3 Honest boundaries

(1) The data reach only $T=6\times10^5$ and $\lambda_{eff}\leq31.4$; "full scale as $\lambda(T)\to\infty$ at infinite height" remains hypothesis (H2′), not a theorem. (2) The super-GUE repulsion is marginally significant (ζ a0.5 deviation -0.00277, ≈3 times the adjacent segment drift 0.00092); it is not used as a theorem hypothesis, only recorded as directionally favorable. (3) The XLI old-convention numbers (global δ, the ρ_L with -1, the $\lambda_{eff}\approx0.5$ annotation) differ from this note; conclusions do not flip but must be disclosed in the XLI revision. (4) The kernel scan is finite (fejer family), not an exhaustive search of the variational space.

5 Next Steps: the S1 Split

5.1 S1a: arithmetic-term attribution (quantifying layer (iii))

(1) Parseval: translate $A(a)-GUE2(a)$ into $F(\alpha,T)-|\alpha|$ for $\alpha\in(0,1)$, compare with the excess above Montgomery's lower bound, and verify sign and shape; (2) decay law: fit $\Delta(a;T)\sim c(a)/\log T$ or $c(a)/\sqrt{\log T}$ over the five segments to distinguish the BGS error structure; (3) kernel-dependent constants: order the deviations of the fejer/sinc/gauss families by the smoothness of their Fourier support (audit6 data already available).

5.2 S1b: construction of the $\lambda(T)$ scheme

Make (H2′) explicit (a strong Montgomery-type statement as $\lambda(T)\to\infty$, with growth rate and error shape); derive $A=1+O(\lambda^{-1})$; incorporate the structural-multiple-zero term ($s<0.01$ already excluded); weighted union → the theorem form with online proportion $1-o(1)$.

References

[1] Lamzouri Y. A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line. arXiv:2609.02882v2 (Proposition 2.1 ($N_0^{\pm}\geq2\Sigma1-\Sigma K^2$), Lemma 3.2 (optimal kernel $f_0(x)=\cos(\sqrt{2}x)/(\sqrt{2}\sin(1/\sqrt{2}))$, $C_{MT}=1/2+(1/\sqrt{2})\cot(1/\sqrt{2})=1.3274992963$), Remark 3.4 ($C_\eta\geq C_{MT}$ optimality)) — read in full.

[2] Baluyot D, Goldston D, Suriajaya A I, et al. An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function. arXiv:2306.04799 (Lemma 5: unconditional pair-

correlation formula, error O_f(1/√logT), weight w(z)=4/(4−z²); the BGS generalization to fixed-modulus L(χ) is an open construction).

[3] Özlük A E. On the pair correlation of zeros of Dirichlet L-functions (1982; at least 11/12 simple under GRH, with weight) — verified via the introduction of Kandhil–Languasco–Moree.

[4] Kandhil N, Languasco A, Moree P. Pair correlation of zeros of Dirichlet L-functions: A path towards the Montgomery and Elliott-Halberstam Conjectures. arXiv:2411.19762 (Yıldırım theorem / Conjecture 1: cross-character-average pair correlation, GRH premise) — read in full.

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[6] Odlyzko A M. The 10^22-nd zero of the Riemann zeta function (10^{10} zeros; numerical verification of RH and GUE pair correlation — “confirmed the GUE connection”).

[7] Defense-XLII numerical package (this work; reproducible with Anaconda python): qrh_work/xlii_s0.py (v1 global convention), xlii_s0_v2.py (v2 local, with the erroneous L density), xlii_s0_v3.py (final: local normalization + two-sided GUE + L calibration), xlii_s0_check.py (Poisson simulation / ρ_2 debugging), xlii_s0_debug.py (minimal verification of the factor 2), xlii_s0_gue.py (GUE simulation), xlii_s0_plot.py (figures); results xlii_s0_v3.json, xlii_s0_gue_sim.json; zero data zeros_T6e5.npz (T=6×10^5, ζ 999354 / L 1149836 roots).

[8] Offense XLI (delivered to d1): conditional form of the pseudorandom union theorem — 7/8 threshold, fejer_a0.5 optimal kernel, decomposition argument, three-layer premise reliability.

[9] Offense XL (delivered to d1): theoretical construction of the unconditional bound on the pairing side — the 0.6725 constant gap is of QRH-mechanism type; Offense XXXIX (S4): fine form of the online-proportion-bridge remainder.